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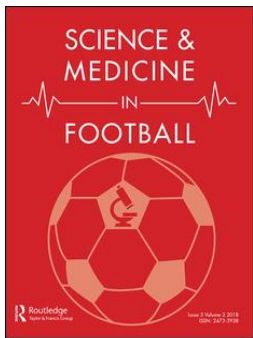


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Characteristics of lower limb force exertion during diving motions by collegiate male soccer goalkeepers

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ABSTRACT

Introduction: Diving is a high-intensity movement with a determining effect on the soccer game. So, the aim of this study was to evaluate the effects of distance and height on lower limb kinetics and ground reaction force (GRF) during a diving motion of soccer goalkeepers.

Methods: The analysis was based on whole-body three-dimensional motion capture (250 Hz) with GRFs recorded using force plates (1000 Hz). Lower limb joint torque and power profile were computed using standard inverse dynamics equations.

Results: Significant effects of both dive distance and ball height were identified based on the magnitude and orientation of the GRF vector for the ball-side leg (BS leg) and on the magnitude of the GRF vector for the contralateral-side leg (CS leg). Overall, flexion torques were negligible, with significant contributions of the hip and ankle to extension torques and power. Of note, the BS leg provided an important negative (braking) power, which increased as a function of increasing ball distance and height.

Conclusion: From these results, spatial parameters of the goalkeeper's diving motion appear to affect the kinetics of the BS leg.

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KEYWORDS

Soccer goalkeeper; diving motion; kinetics

Introduction

Despite their overall low occurrence, goalkeeper (GK) saves (i.e. preventing goals) (9.96 ± 3.8 per match, De Baranda et al. 2008; GK saves of high, intermediate, and low level teams: 2.9 ± 1.8 , 3.7 ± 1.9 , and 3.4 ± 2.1 per match, respectively; Liu et al. 2015) have been found to be the most influential contribution of GKs to the outcome of a soccer game (White et al. 2018). Essentially, training for GKs should include a combination of high-intensity movements and technical elements. Diving is a high-intensity movement with a determining effect on the game. When GKs have insufficient time to move the feet to align with the shoot, they dive towards it (Welsh 2014). The dive of a GK has a feature that is not seen in other jumping motions, such as jumping sideways toward a ball that is approaching from the front. In practice, during the 2002 FIFA World Cup in Korea and Japan, GKs performed up to 17 dives (6.2 ± 2.7) per match (De Baranda et al. 2008). A study, Matsukura and Asai (2013a) and Knoop et al. (2013) reported that GKs require ~ 1.3 s to recognise the course of the ball, reach the edge of the goal, and arrive at the ball. Moreover, different biomechanical strategies can be used to perform the combinations of movements underlying the diving motion, based on the ball height and flight distance (Graham-Smith et al. 1999). To be effective, strategies need to produce large ground reaction forces (GRFs) to optimise the velocity and linear motion of the body's centre of gravity (CG) towards the ball (Suzuki et al. 1987; Spratford et al. 2009; Elleray 2013). Highly-skilled GKs generated a larger centre of gravity velocities (CGV) upon take-off (Suzuki et al. 1987). Furthermore,

compared with the non-preferred side, higher CGVs were observed when diving on the preferred side (Spratford et al. 2009). Based on the findings of Spratford et al. (2009), the effects of strength asymmetry on strategies, including greater external rotation of the pelvis and chest in combination with higher knee extension torque, lower ankle plantarflexion torque, and excessive movement of the CG, are observed during the initial phase of a diving motion to the non-preferred side. The above-mentioned studies indicate that executing a superior dive requires the generation of a large amount of joint torque, in order to produce GRFs that will cause the CG to move in the correct direction in a very short period of time. However, there are no reports of quantitative analysis of strategies employed to adapt the GRFs, and associated lower-limb joint torques, to the path and flight of the ball. Furthermore, with regard to the long jump, which is a jumping event, the importance of force exertion mechanisms at the time of take-off has been highlighted, and training suggestions were noted from the findings (Muraki et al. 2008). Hence, elucidating the force exertion mechanisms at the time of take-off may greatly contribute to the training and performance improvement of GKs. In actual game situations, it is necessary to perform diving motions with various courses and speeds of the ball and shot timings, but as the factors affecting the motion increase, conducting experiments under conditions that consider all factors is deemed difficult. Therefore, as a first step towards analysing GKs' diving motion, our study aimed to use three-dimensional motion analysis to characterise profiles of

lower-limb joint torques and power, based on GRF data, and evaluate the effects of the course and flight of the ball on lower limb kinetic and GRF profiles. It was hypothesised that GRF data and lower-limb joint torques and power would change during diving motion according to the differences in distance and height of the suspended ball.

Methods

Participants

Eleven male university football GKs were recruited into our study group, with a mean (standard deviation) height of 1.78 (0.06) m and mass of 71.2 (2.5) kg. All participants had played as GKs for 9.7 (3.1) years and were able to perform a skilled diving manoeuvre. Participants provided informed consent, and our study was approved by the research ethics board of our institution.

Motion capture and ground reaction force

Movements were recorded using the VICON 3D motion capture system (VICON 612, 250 Hz, nine cameras, Oxford Metrics, Oxford UK). For motion capture, 47 reflective markers were affixed to the body (Suzuki et al. 2014), and the motion of the marker was recorded at 250 Hz. The GRFs were recorded using two force platforms (9281A, platform, 90 cm square, Kistler Corporation, Winterthur, Switzerland and 9287B, platform 60 cm square, Kistler Corporation) at a sampling rate of 1000 Hz. These two force platforms with different sizes were from the same manufacturer. The difference in size did not influence the outcome measures and GKs' motions during the experimental trial. For analysis, the global coordinate system was defined, with the x-axis oriented along the plane of forward-backward movement of the GK (positive x forward), the y-axis along the plane of right-left horizontal movement (positive y to the right), and the z-axis along the plane of vertical movement (positive z upward).

Experimental diving manoeuvres

Diving motions were analysed for two ball distances (near and far) and three ball heights (upper, middle, and lower), with six experimental conditions tested: near-distance/upper ball (NU), near-distance/middle ball (NM), near-distance/low ball (NL), far-distance/upper ball (FU), far-distance/middle ball (FM), and far-distance/low ball (FL). All balls were presented to the GK's right side only. Movements towards the ball were standardised between GKs, based on the movement strategies and depending on ball distance and height, as described by Graham-Smith et al. (1999). For all near-distance balls, the GK was positioned at the centre of two force plates, with the ball placed 1.83 m away. For all far-distance balls, GKs were positioned 1.2 m outside of the force plate, allowing them to take an initial step towards the ball and contact the force plate, and the ball was placed 2.3 m from the centre of the two force plates. This resulted in the ball being at a total distance of 3.5 m from the initial position of the GK (Figure 1). Each GK was then instructed to perform the dive from his initial stance for near-distance balls. For far-distance balls, GKs first took one sidestep towards the ball and then performed the dive. For calculation of kinetic and kinematic data, the experiments were performed indoors; participants wore their own commercially available indoor football shoes. These limitations of the experimental design were considered when choosing the parameters to be assessed in this study.

In accordance with previous studies (Suzuki et al. 1987; Spratford et al. 2009), the ball height used in the experiment was standardised by hanging the balls (using a Pelada TETHER, Molten Corporation, Hiroshima, Japan) on the goal post and placing a normal soccer ball (Pelada PF405, Molten Corporation) on the safety mat. For upper height conditions, the hanging ball was adjusted such that the CG of the ball was 2.2 m above the ground. This was adjusted to 1.22 m for middle-height conditions. For low-height conditions, the normal soccer ball was positioned on a safety mat placed on the ground. These standardised heights were used for all dives.

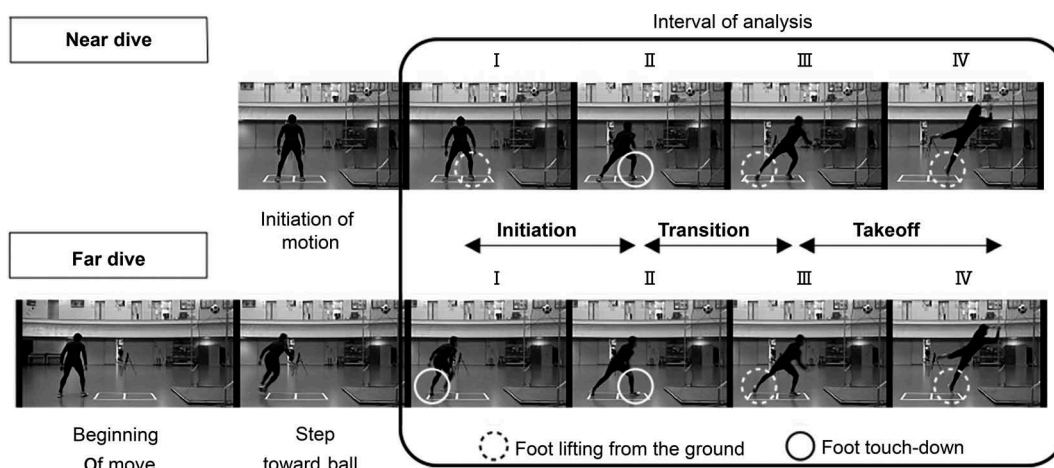


Figure 1. Phases of the diving manoeuvre. The upper panel shows an example of a near dive, and the lower panel shows an example of a far dive. All figures are shown from a posterior view.

Definition of the phases of the diving motion

Here, diving motions were subdivided into three phases so players and coaches could recognise and analyse them during training and game situations: Initiation, Transition, and Takeoff. These phases were defined by four-time points (I–IV). *Start of analysis* (I) was defined for near-distance dives by the lifting of the ball-side (BS) leg from the ground, and for far-distance dives, by the touchdown of the contralateral-side (CS) leg. The end of the *Initiation* phase (and, accordingly, the start of the *Transition* phase [II]) was defined by the touchdown of the BS foot. The end of the *Transition* phase (and, accordingly, the start of the *Takeoff* phase [III]) was defined by the lifting of the CS leg from the ground. The *endpoint of analysis* (IV) was defined by take-off of the BS leg.

Measured parameters for analysis

The CGV was evaluated using body segment inertial parameters according to Ae (1996) to calculate the CG and velocity of the CG displacement.

The GRF was divided by mass for between-subject comparison (N/kg). The BS leg was not in contact with the force plate during the *Initiation* phase and the CS leg was not in contact with the force plate during the *Takeoff* phase. From the GRF data, we calculated the magnitudes and angles of horizontal and vertical projection vectors. All vector data were normalised to 100% from the start to endpoint of the analysis.

In terms of kinematics, angular velocity computed as the first derivative of movement of a segment and the relative position of each body segment were calculated. The joint angle and angular velocity were calculated using the moving coordinate system (Matsukura and Asai 2013b) and were defined for the lower torso and hip, knee, and ankle joints. Joint angular velocity was calculated by differentiating the joint angle with respect to the axes of hip flexion and extension, hip adduction and abduction, knee flexion and extension, and ankle dorsiflexion. More detailed information on these calculations is provided in the Supplementary Material (Supplement 1). Joint torque and power were calculated using inverse dynamics equations, based on the Newton-Euler method. In addition, to determine the angular velocity vector for each body segment, we defined a moving coordinate system that indicated a pose for each body segment and calculated the angular velocity vector from the time derivative of each vector component of this system. In the moving coordinate system of each segment, it was assumed that the lower torso was consistent with the lower torso coordinate system, the thigh with the hip internal-external rotation coordinate system, the lower leg with the knee internal-external rotation coordinate system, and the foot with the ankle dorsiflexion coordinate system. The lower torso hip internal-external rotation coordinate systems are described in Supplement 1. More details of the inverse dynamics equations are provided in the Supplementary Material (Supplement 2).

Joint torque powers were calculated as the products of joint torque and joint angular velocities (W). Motion and GRF data that were used as inputs for the inverse dynamic calculation were smoothed using a low-pass Butterworth

filter (5–15 Hz), with the cut-off frequencies determined using the Wells and Winter method (1980). The scalar product of the unit vectors along all three planes of motion (sagittal, medial-lateral, and superior-inferior) was calculated to obtain the total joint torque around each joint axis, with the torque divided by GK mass to allow between-subject comparisons (Nm/kg).

Statistical analysis

Descriptive statistics, including waveforms, were calculated for each outcome variable: horizontal and vertical CGV, joint torque, and power. The effects of ball height and distance on joint torque and power profiles were evaluated using a two-way factorial analysis of variance (ANOVA), with Bonferroni correction for multiple comparisons. Estimates of the mean were also calculated as appropriate. All analyses were performed using SPSS (version 24, IBM SPSS Inc., Armonk, NY, USA), with the level of significance set at 0.05, and the 95% confidence intervals of the mean and effect size of ANOVAs (partial η^2) were calculated.

Results

Effects of ball distance and height on the ground reaction force profile

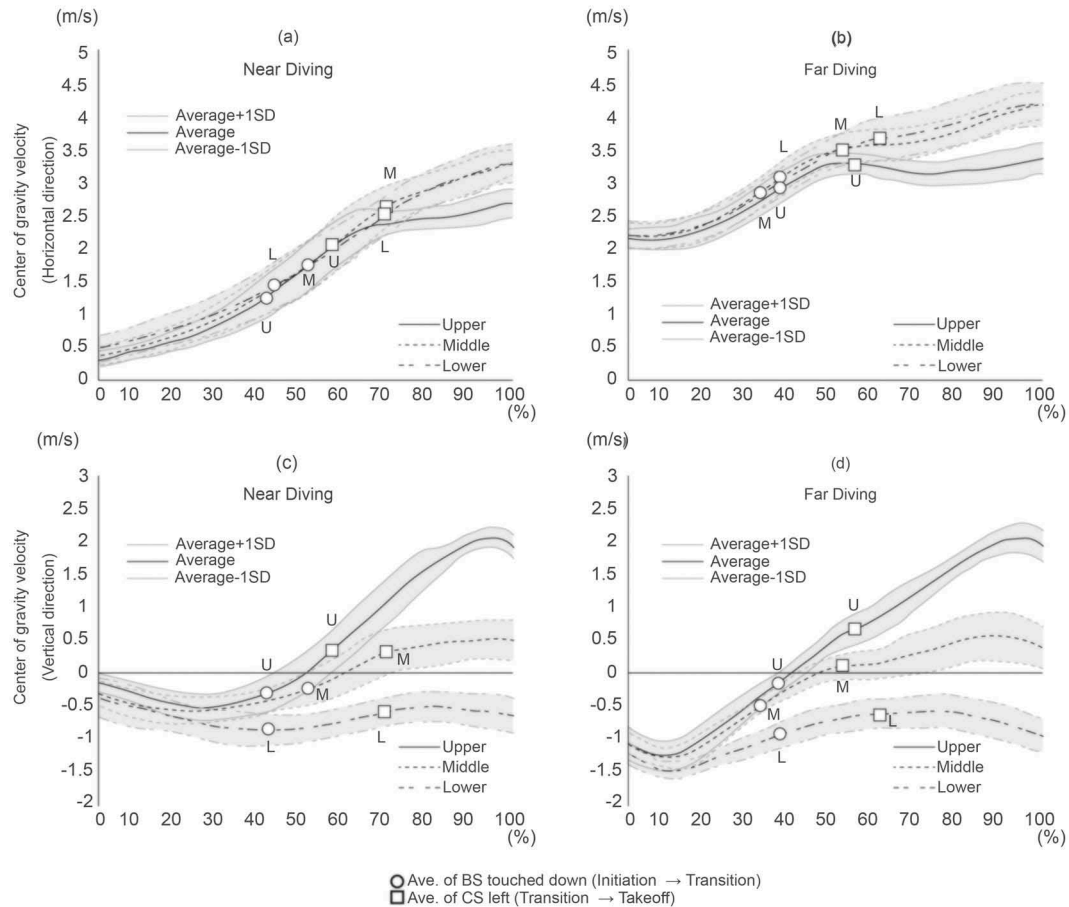
Table 1 shows the CGV for each time point in each trial, with the statistical analysis of the raw data summarised in Supplementary Table 1. Far diving showed a significantly larger CGV value in the horizontal plane at the end of Take off phase (endpoint of analysis, IV), and diving for low ball positions generated significantly larger CGV values than the other heights. Contrarily, in the vertical plane, the CGV values recorded for both near and far diving increased significantly as the height of the ball increased. There were no significant differences between the CGV values for near and far diving except in the case of low ball heights.

The average CGV profile is shown in Figure 2 and was normalised to 100% from the start to end of the analysis. As the figure illustrates, the horizontal CGV increased from the start of the analysis towards the end. The acceleration decreased close to BS touchdown. The decrease in acceleration was particularly noticeable during the upper diving motion (Figure 2(a,b)). In the vertical plane, each diving motion showed a decreasing CGV until BS touchdown. Thereafter, the CGV changed to an upward direction at a time point close to BS touchdown, during dives for upper- and middle-height balls. The downward velocity decreased during dive for lower-height balls (Figure 2(c,d)). The horizontal CGV for the BS leg decreased with increasing vertical CGV. The peak vertical CGV increased as a function of increasing ball height.

Figure 3 shows a stick figure representation, posterior view, of a dive to the right for the representative GK whose height and mass were close to the mean value of the study group. Magnitudes of synthetic vectors for the horizontal and vertical GRFs are also shown at 10% intervals from the start to end of motion analysis. For the CS leg, vector magnitudes

Table 1. Means of the centre of gravity velocity (CGV, m/s) at each time point. Abbreviations: FL, far lower; FM, far middle; FU, far upper; NL, near lower; NM, near middle; NU, near upper.

		Initiation		Transition		Takeoff	
		Start		End/Start		End/Start	End
		I		II		III	IV
Vertical CGV (m/s)	NU	† 0.33 (0.11)	*]	† 1.58 (0.44)	*]	† 2.35 (0.27)	† 2.67 (0.20)
	NM	† 0.38 (0.13)		† 1.67 (0.37)		† 2.76 (0.27)	† 3.14 (0.45)
	NL	† 0.47 (0.20)		† 1.44 (0.37)		† 2.91 (0.33)	† 3.25 (0.29)
	FU	† 2.14 (0.15)	*]	† 2.91 (0.31)	*]	† 3.29 (0.14)	† 3.35 (0.22)
	FM	† 2.21 (0.18)		† 2.97 (0.39)		† 3.56 (0.25)	† 4.15 (0.21)
	FL	† 2.21 (0.22)		† 2.80 (0.27)		† 3.68 (0.31)	† 4.15 (0.33)
Horizontal CGV (m/s)	NU	† -0.14 (0.15)	*]	-0.06 (0.33)	*]	† 1.05 (0.67)	1.75 (0.26)
	NM	† -0.30 (0.19)		-0.31 (0.20)		† 0.41 (0.31)	0.47 (0.30)
	NL	† -0.38 (0.30)		-0.90 (0.18)		† -0.48 (0.23)	† -0.65 (0.25)
	FU	† -1.05 (0.24)	*]	-0.14 (0.37)	*]	† 0.78 (0.30)	1.88 (0.26)
	FM	† -1.05 (0.16)		-0.44 (0.35)		† 0.14 (0.20)	† 0.36 (0.31)
	FL	† -1.21 (0.17)		-1.05 (0.22)		† -0.56 (0.21)	† -0.96 (0.25)

* Indicates a significant difference between heights ($P < 0.05$)† Indicates a significant difference between near and far diving ($P < 0.05$)**Figure 2.** Average centre of gravity velocity (CGV, m/s) patterns over different phases of the diving motion from 0% to 100%. The upper two panels (a–b) show the horizontal CGV and the lower two panels (c–d) show the vertical CGV.

increased with increasing ball height, but with a consistent direction of force application at 60° to the right. For the BS leg, vector magnitudes increased with increasing ball height

with a change in angle, toward the ground, 50%–90% of the diving motion for low balls. A minimum angle was observed at 90% of the motion, which was recorded to be 47.4°

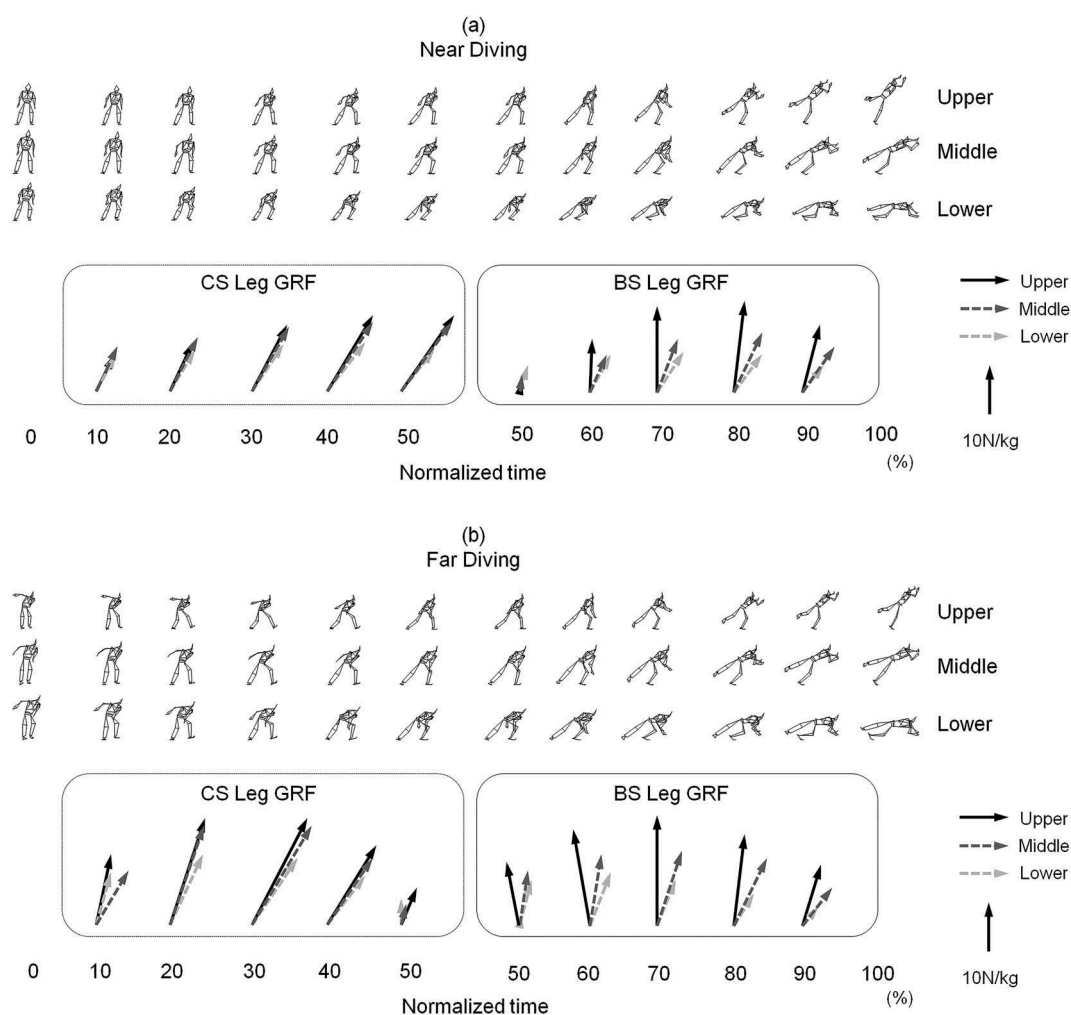


Figure 3. Typical patterns of the diving motions, with stick figures showing the transition in body motion over the entire dive, from 0% to 100%. The arrow shows the vector of the average ground reaction force (GRF). The difference in the slope and magnitude of the GRF vector is clearly evident. Upper figures (a) show the pattern of near diving, and lower figures (b) show the pattern of far diving.

(relative to ground, 0°) for NL and 49.3° for FL. For high balls, the GRF produced in the opposite direction of the diving motion provided a braking force.

Effects of ball distance and height on lower kinetic profiles

The average joint torque and joint torque power profiles of the BS leg during the *Takeoff* phase are shown in Figure 4, with the statistical analysis of the raw data summarised in Supplementary Table 2. As the hip/knee flexion and ankle dorsiflexion torque values were extremely small, only hip/knee extension and ankle plantar flexion torque were analysed. In addition, as the average hip/knee flexion and ankle dorsiflexion powers were also extremely small compared to the extension and plantar flexion powers, the former are not shown on the figures.

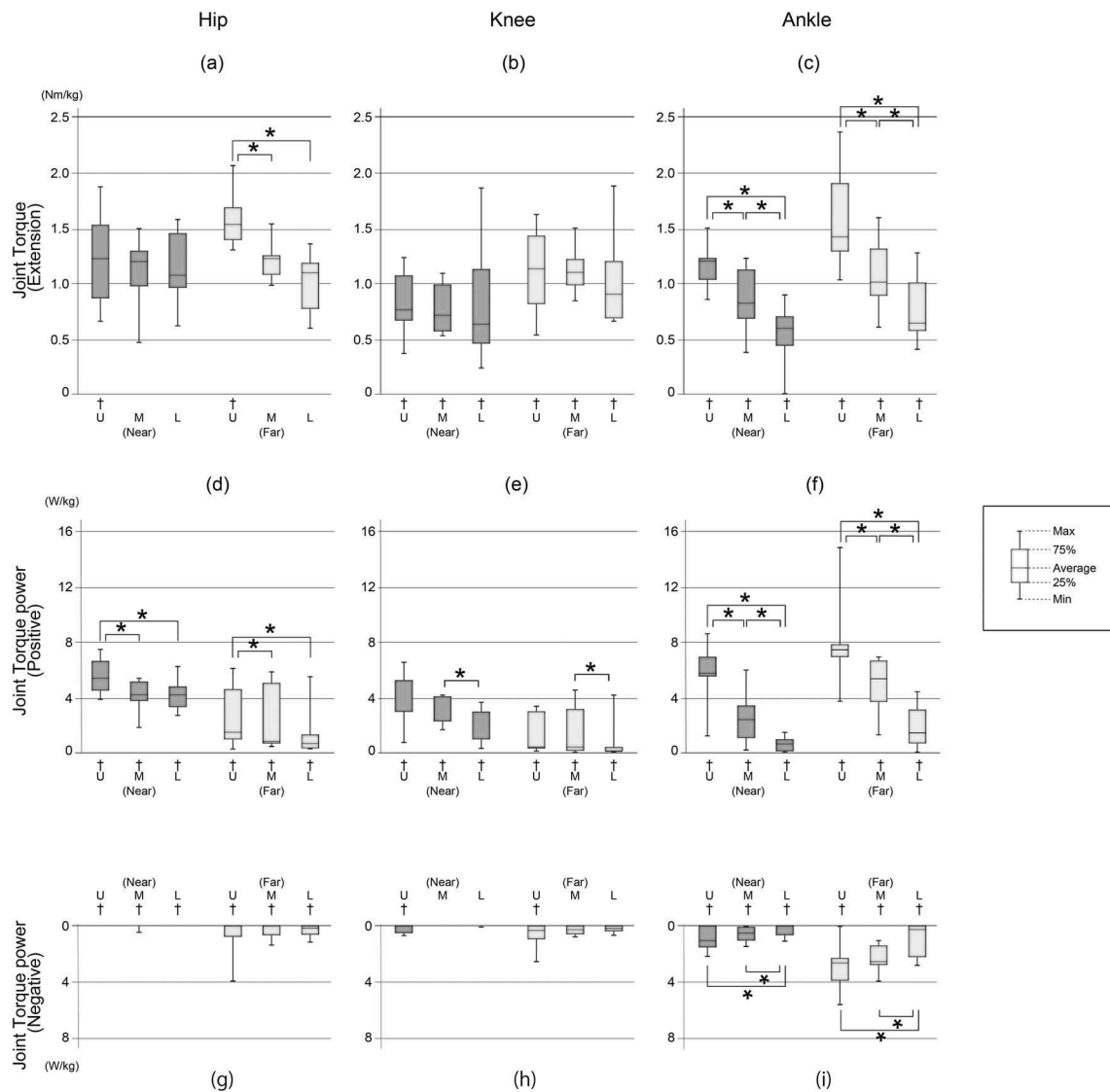
For the BS leg, hip extension torque was greater for upper ball height dives, with knee and ankle torque being significantly greater for all far-distance dives ($P < 0.05$, Figure 4(a–c)). Positive power was significantly greater at the hip and knee for all near-distance dives ($P < 0.05$, Figure 4(d,e)) and at the ankle for all far-distance dives ($P < 0.05$, Figure 4(f)). Furthermore, negative

power was significantly greater at the ankle during the *Takeoff* phase for far-distance dives ($P < 0.05$, Figure 4(i)).

Considering ball height, joint torque and powers were generally higher for upper ball height dives ($P < 0.05$). Extension torques were significantly greater at the hip for FU dives and at the ankle for all upper ball height dives ($P < 0.05$, Figure 4(a–c)). Positive power was significantly greater at the hip and ankle during *Takeoff* phase for all upper ball-height dives ($P < 0.05$, Figure 4(d–f)). Moreover, negative power tended to be greater at the ankle than at the hip and knee joints (Figure 4(g–i)), with the greatest negative ankle power being generated during the *Take off* phase for upper ball height dives ($P < 0.05$, Figure 4(i)).

Discussion

The primary result of this study is the demonstration that CGV at the time of take-off (Table 1) is affected by the distance between the GK and the ball and the height of the ball. At all ball heights, far diving was associated with a significantly larger horizontal CGV than near diving. In far diving, this parameter is thought to be strongly influenced by the CGV, which is acquired in the step taken before the *start of analysis* phase.



* Indicates a significant difference between heights ($P<0.05$).
† Indicates a significant difference between near and far diving ($P<0.05$)

Figure 4. Ball-side leg mean joint torques (Nms/kg) and positive and negative joint torque power (W/kg) during the *Takeoff* phase, normalised to body mass when the torque was produced in extension or plantarflexion among GKs. Upper panels (a–c) show mean joint torques, middle panels (d–f) show mean positive joint torque power, and lower panels (g–i) show mean negative joint torque power. Left panel images (a, d, and g) relate to hip extension, centre panel images (b, e, and h) relate to knee extension, and right panel images (c, f, and i) relate to ankle plantarflexion. Abbreviations: U, upper; M, middle; L, lower; GK, goalkeeper.

Furthermore, at upper ball height, both near and far diving had significantly larger vertical CGVs than at lower ball height. It is therefore thought that during the *Takeoff* phase of the diving motion, the GK adjusts the GRF depending on the distance to the ball and the height of the ball in order to dive towards the ball at appropriate angles and speeds. This study also provides evidence of the adjustment of GRF of the BS leg to both ball distance and height, and to the ball distance for only the CS leg. This asymmetrical control of GRF appears to be important in order to achieve linear movement of the body's CG towards the ball. Around BS touchdown, deceleration in the horizontal CGV that was observed for high balls was consistent with temporal increases in the magnitude of the vertical CGV (Figure 2(a,b)). This negative horizontal power of the BS leg would provide an

important contribution to breaking the forward motion of the CG during the *Transition* to *Takeoff* phase. Therefore, this could contribute specifically to the change in orientation of the vertical GRF under the BS leg in the case of a higher ball; a change that is not observed in the CS leg (Figure 3(a,b)). For the CS leg, vector magnitudes increased with increasing ball height with a consistent direction of force application at 60° to the right. When we further consider the angle of the GRF of the BS leg at an elevation <50° relative to the ground (Figure 3), the extension power that is generated as the BS leg contact to the ground is likely to be organised so as to decrease the downward CGV (Figure 2(c,d)). This would explain the fact that dives for low balls are not linearly aligned with the ball and take longer to execute than dives for high balls (Matsukura and Asai

2013a). These findings indicate that the CS leg adjusts the magnitude of the force after the BS leg adjusts both the magnitude and direction of the force to the height of the ball.

Overall, for the BS leg, a greater average joint torque was produced for far and upper dives than for near and lower dives (Figure 4(a,c)). Meanwhile, the magnitudes of the positive joint torque power of the BS leg at the hip and knee were greater for near- than for far-distance dives and that at the hip was greater for upper- than for middle or lower dives during *Takeoffs* (Figure 4(d,e)). While positive joint torque power was calculated as the inner product of the extension torque and the extension angular velocity at a joint, we speculate that the higher torque power calculated for the BS leg with near-distance dives reflects a higher angular velocity of the hip and knee joint during the diving motion. As GKs did not take a step before near-distance dives, the BS leg would not have a large amount of kinetic energy that would be readily available to increase the velocity of extension movement and produce a large positive torque power. Contrarily, in far diving, although the joint torque power was small, the GRF vector at 50% and 60% points for the BS leg are larger than for near diving (Figure 3). As kinetic energy for far diving is generated from the step, large torque power is not necessary. Therefore, we propose that joint torque is larger during far diving than during near diving in order to maintain posture against this large GRF and prevent collapse. Furthermore, the average positive and negative joint torque power produced by the BS ankle was greater for far-distance and upper dives compared to near-distance and lower dives (Figure 4(f,i)). During take-offs for running jumps, the negative power produced by the ankle absorbs energy, produces a large amount of positive power, which helps to increase the velocity of the CG in the vertical and horizontal directions (Muraki et al. 2008). We suggest that the exertion of both positive and negative powers by the ankle demonstrates the plyometric response of the ankle plantar flexor muscles.

In the case of diving for balls that were near to the GK (NM, NL), we expect that the GK reached the ball without completely extending the ankle joint. As a result, it can be assumed that positive power was significantly greater at the ankle for all far-distance dives.

Finally, similar to a previous study (Spratford et al. 2009), the limitation here was the use of a stationary hanging ball. This configuration was required to enable a consistent distance and height from the GK. Therefore, it concerns that the movement pattern of the upper body towards the ball and the cognitive task of predicting the ball trajectory will be different from that in an actual game. Future research will need to focus on change in these basic strategies to the dynamic context of ball motion.

Conclusions

Using three-dimensional motion capture, we identified unique biomechanical strategies of the BS leg over three phases of diving motion. Our results explain the differences in the ability to reach near, far, high, and low balls. We provided evidence on the asymmetric control of GRF where the GK adjusts the magnitude of the GRF with each leg, however, adjusts the orientation of the GRF with only the BS leg.

We identified the dominant contribution of hip extension torque power to the performance of near-distance dives, with the ankle plantar flexion torque playing a dominant role in far-distance dives. This indicates the flexibility of the underlying biomechanics based on kinematic features of the ball. The ankle plantar flexion torque also contributed an important horizontal braking force, with the negative power increasing during dives for high balls, which require optimisation of the vertical component of the GRF. Generally, at take-off, the bilateral extension power is important with the magnitude of the extension power positively correlated with ball distance. Focus on these key biomechanical lower limb components could improve the diving performance of GKs.

Practical implications

The results of this study provide important information regarding the general principles of biomechanical organisation explaining the differences in dive performance that are observed with respect to ball distance and height. Near-distance dives without an approach step, especially NU dives, require a rapid hip extension and generation of large positive power. When diving for far distance or high balls, ankle plantar flexors contribute a large positive power allowing the GK to arrive at the ball quickly. The energy absorption capacity of the ankle plantar flexors is more important during diving for high balls than for low balls. Therefore, training programs facilitating these specific components (plyometric training, which improves stretch-shortening cycle performance [Hoffman 2014]) could improve the diving performance of GKs.

Disclosure statement

No potential conflict of interest was reported by the authors.

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